 # Simple Line Plot

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,30,40,50]

plt.plot(x, y)
plt.title("Simple Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

 # Plot Student Marks

```
import matplotlib.pyplot as plt

students = ["A","B","C","D","E"]
marks = [75,85,60,90,70]

plt.plot(students, marks, marker='o', color='green')
plt.title("Student Marks Line Plot")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()
```

 # Bar Chart

```
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English", "CS"]
marks = [80,75,90,85]

plt.bar(subjects, marks)
plt.title("Subject Marks")
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.show()
```

 # Scatter Plot

```
import matplotlib.pyplot as plt

height = [150,160,165,170,175,180]
weight = [50,55,60,65,70,75]

plt.scatter(height, weight)
plt.title("Height vs Weight")
plt.xlabel("Height")
plt.ylabel("Weight")
plt.show()
```

 # Histogram

```
import matplotlib.pyplot as plt

marks = [55,60,65,70,75,80,85,90,95,60,70,80]

plt.hist(marks)
plt.title("Histogram of Marks")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()
```

 # Pie Chart

```
import matplotlib.pyplot as plt

languages = ["Python", "Java", "C++", "JavaScript"]
students = [40,25,20,15]

explode = (0.1, 0, 0, 0)

plt.pie(students, labels=languages, explode=explode, shadow=True, autopct='%1.1f%%')
plt.title("Programming Language Preference")
plt.show()
```

 # Customize Plot

```
import numpy as np
import matplotlib.pyplot as plt

x = np.arange(1, 6)
y = x * 2

plt.plot(x, y, marker='o', linestyle='--', color='red')
plt.title("Customized Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

 # Plot Sales Data

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May"]
sales = [200,300,250,400,350]

plt.plot(months, sales, marker='o')
plt.title("Monthly Sales Data")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

Outputs:

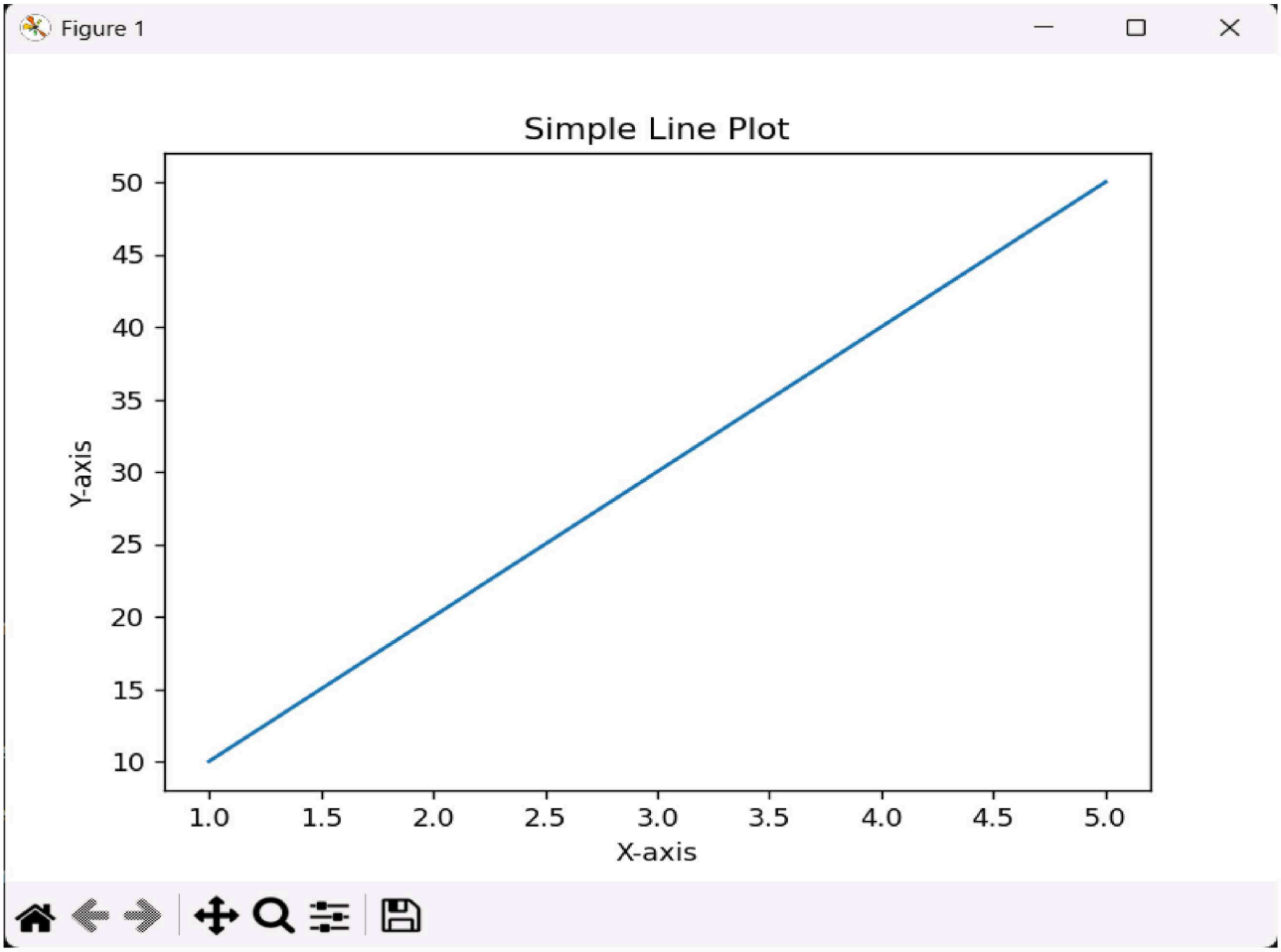
Figure 1
Simple Line Plot
Y-axis
X-axis
50
45
40
35
30
25
20
15
10
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

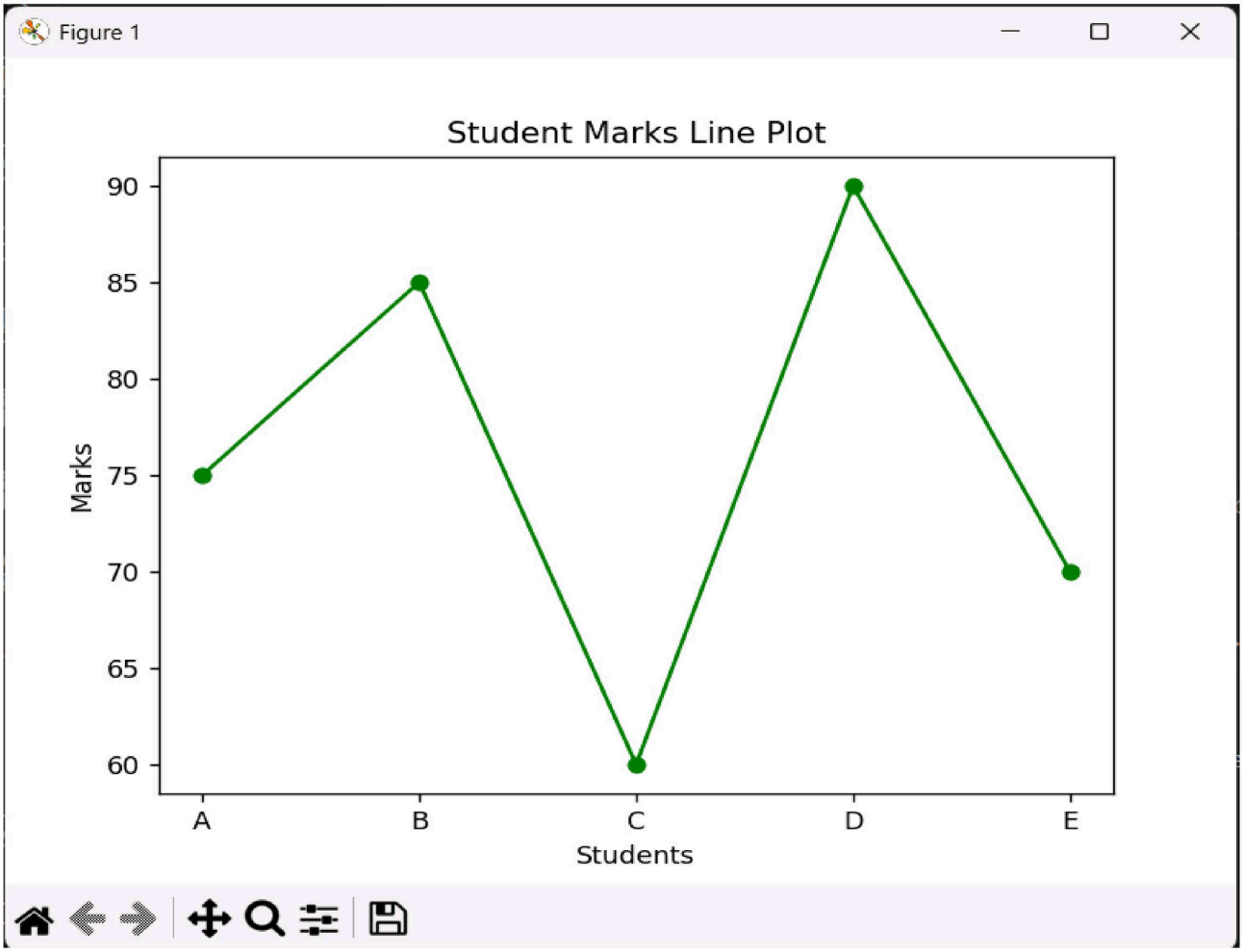
Figure 1
Student Marks Line Plot
Marks
Students
90
85
80
75
70
65
60
A
B
C
D
E

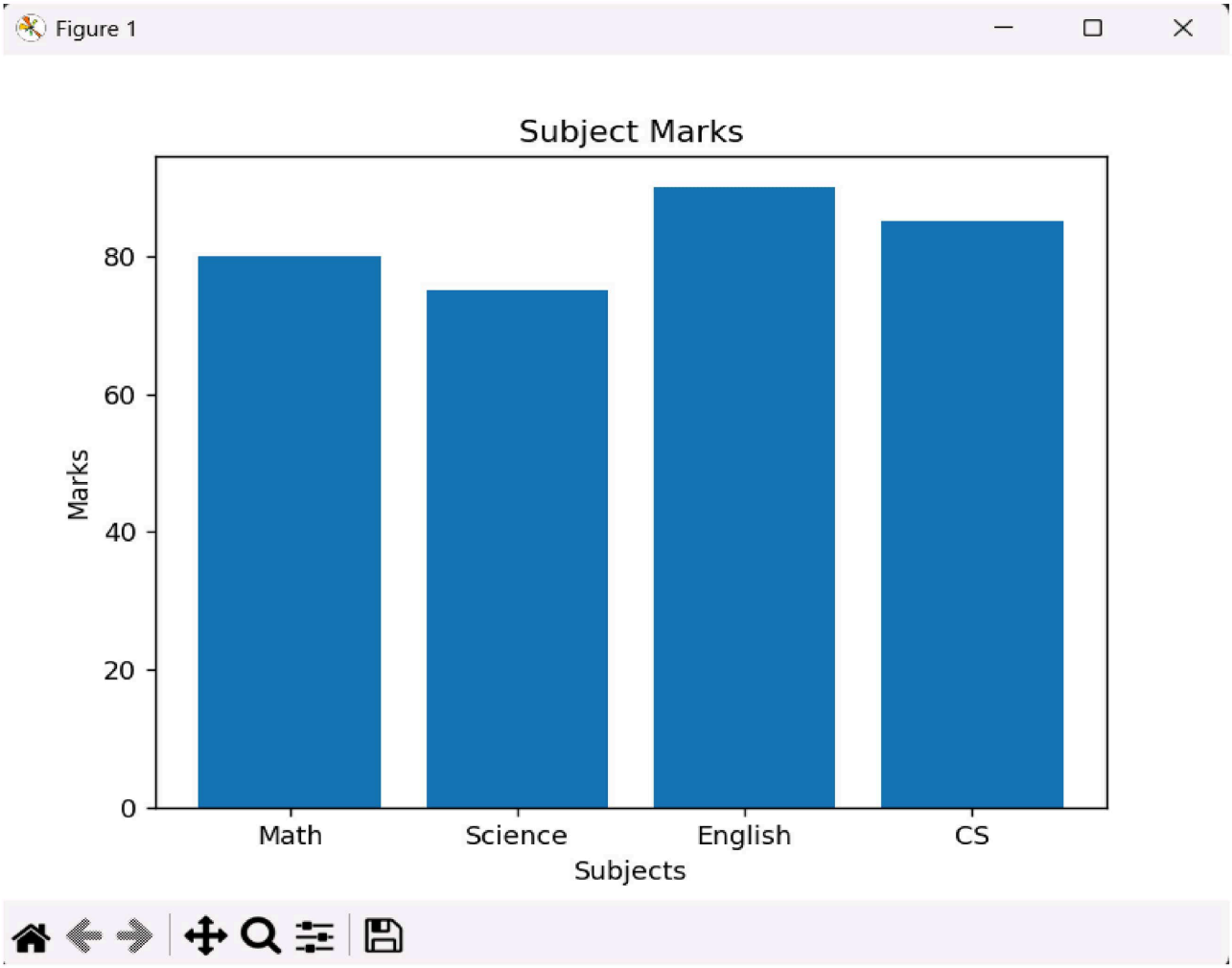
Figure 1
Subject Marks
Marks
0
20
40
60
80
Math
Science
English
CS
Subjects

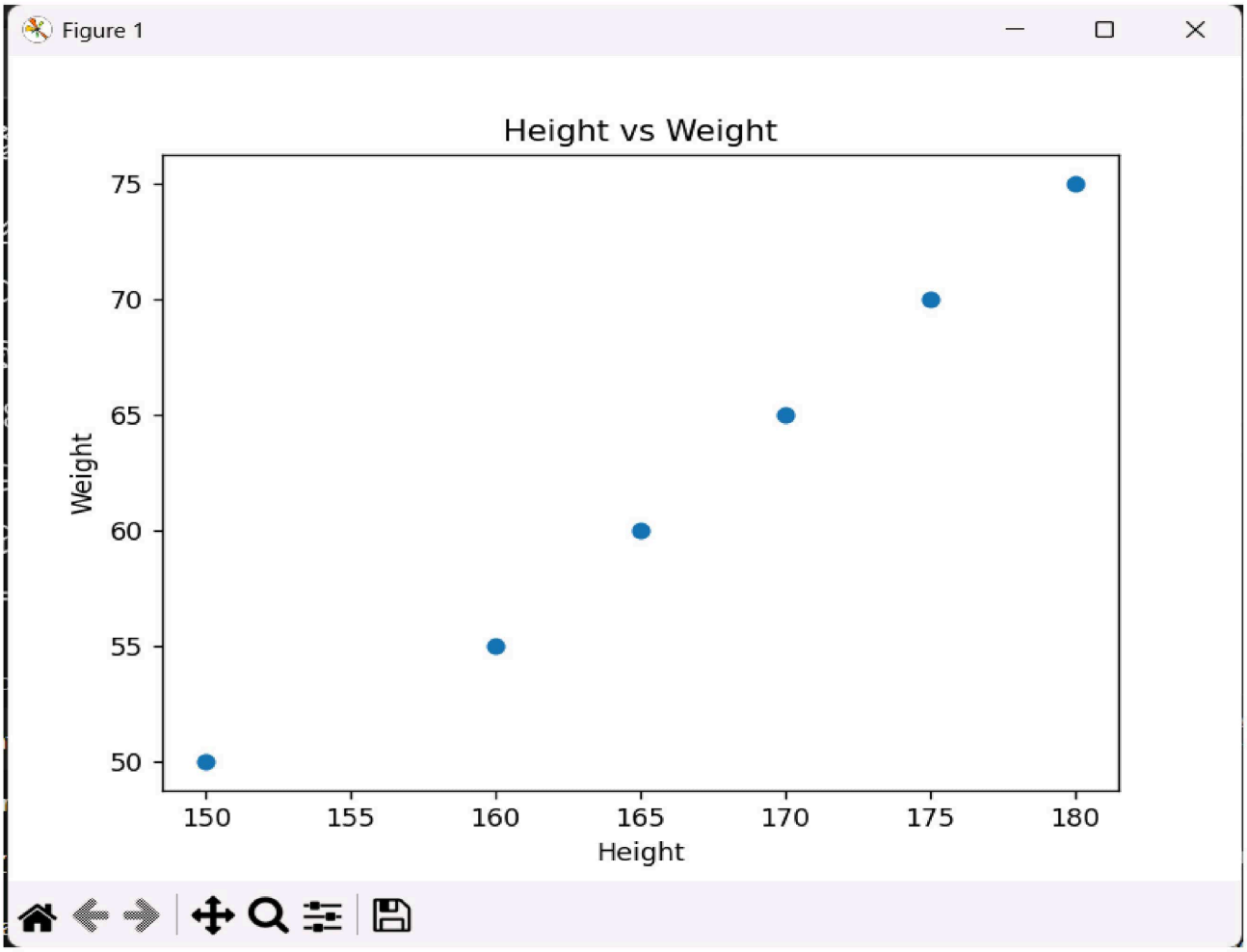
Figure 1
Height vs Weight
Weight
50
55
60
65
70
75
150
155
160
165
170
175
180
Height

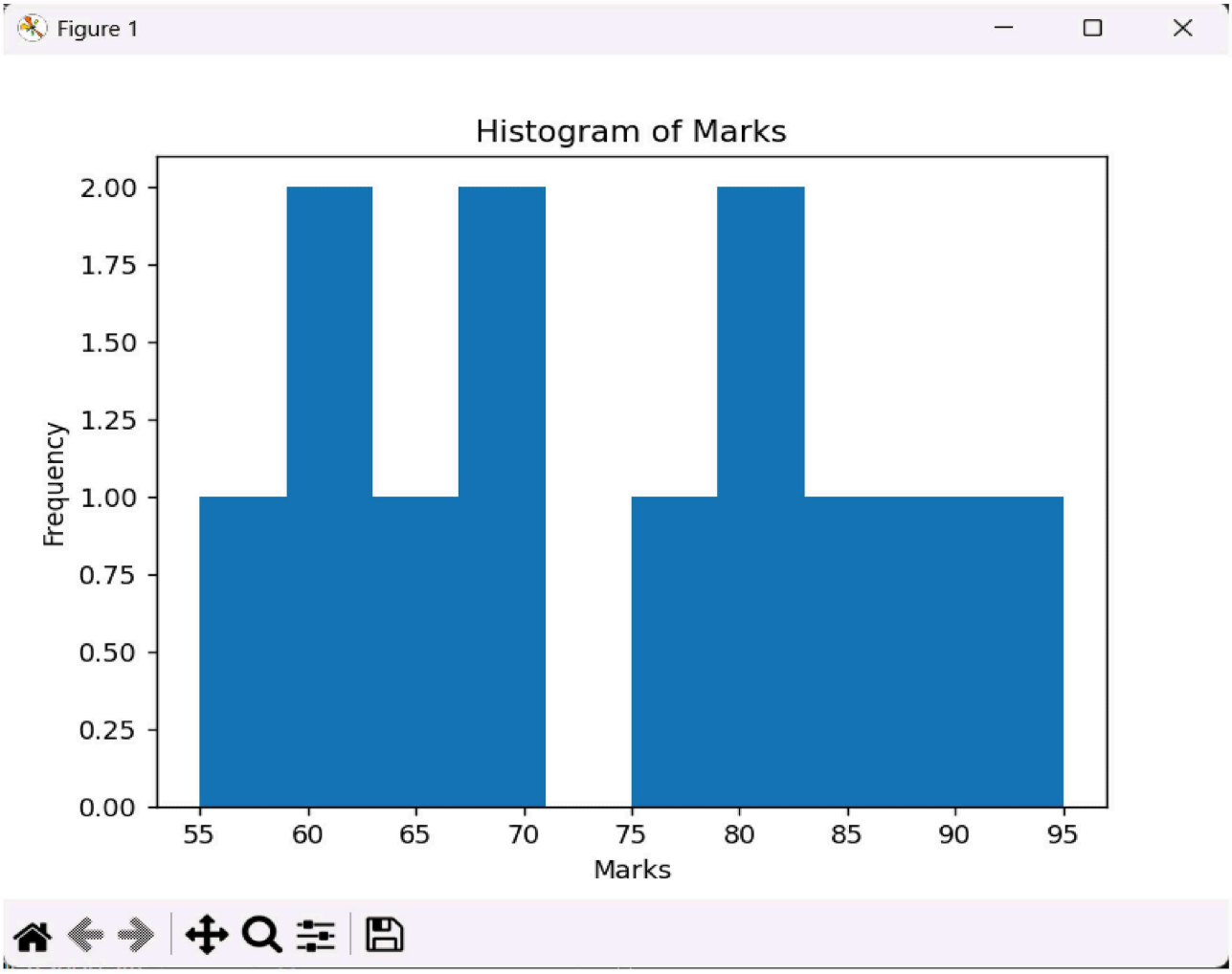
Figure 1
Histogram of Marks
Frequency
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
55
60
65
70
75
80
85
90
95
Marks

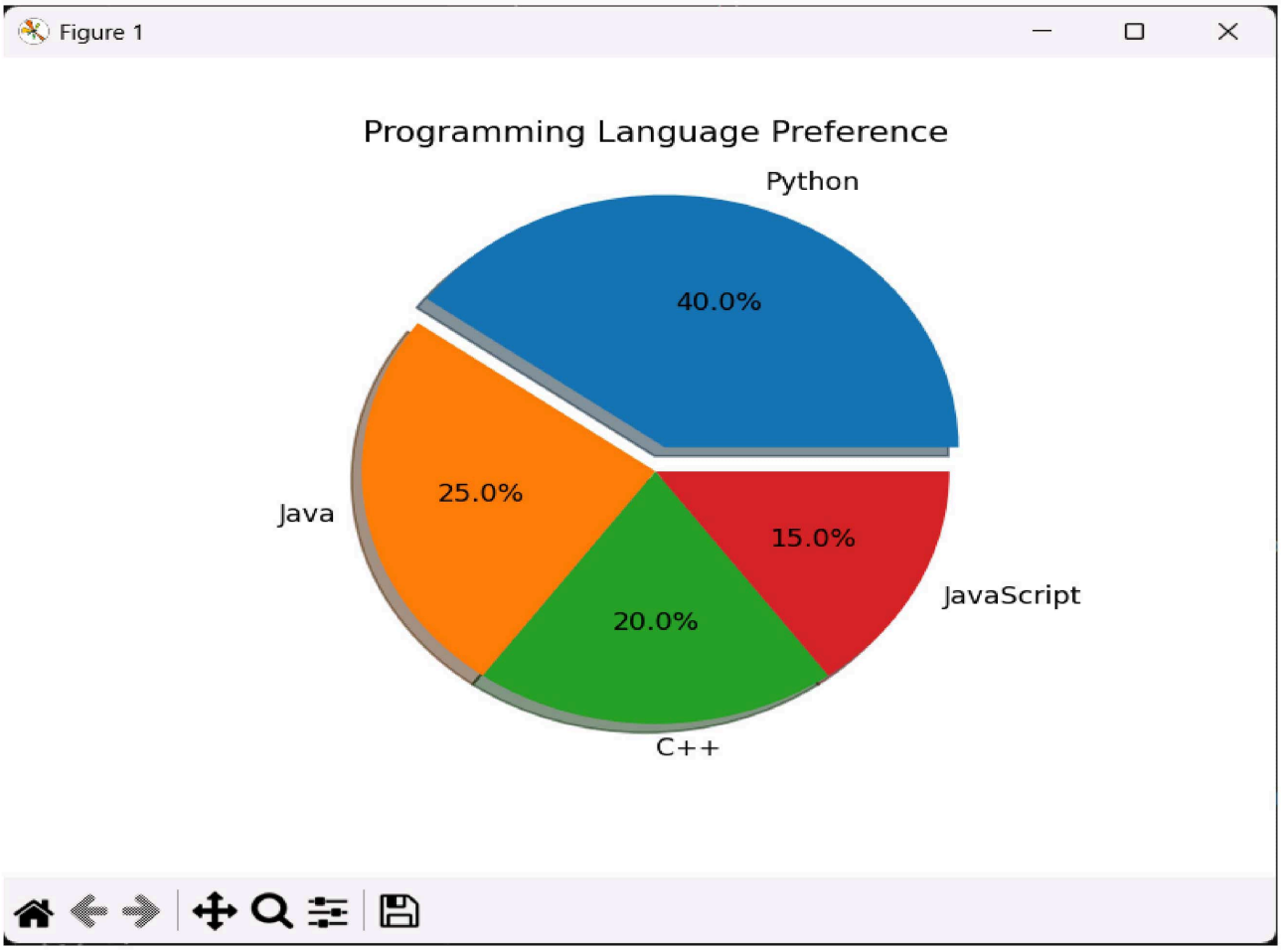
Figure 1
Programming Language Preference
Python
40.0%
Java
25.0%
15.0%
JavaScript
20.0%
C++

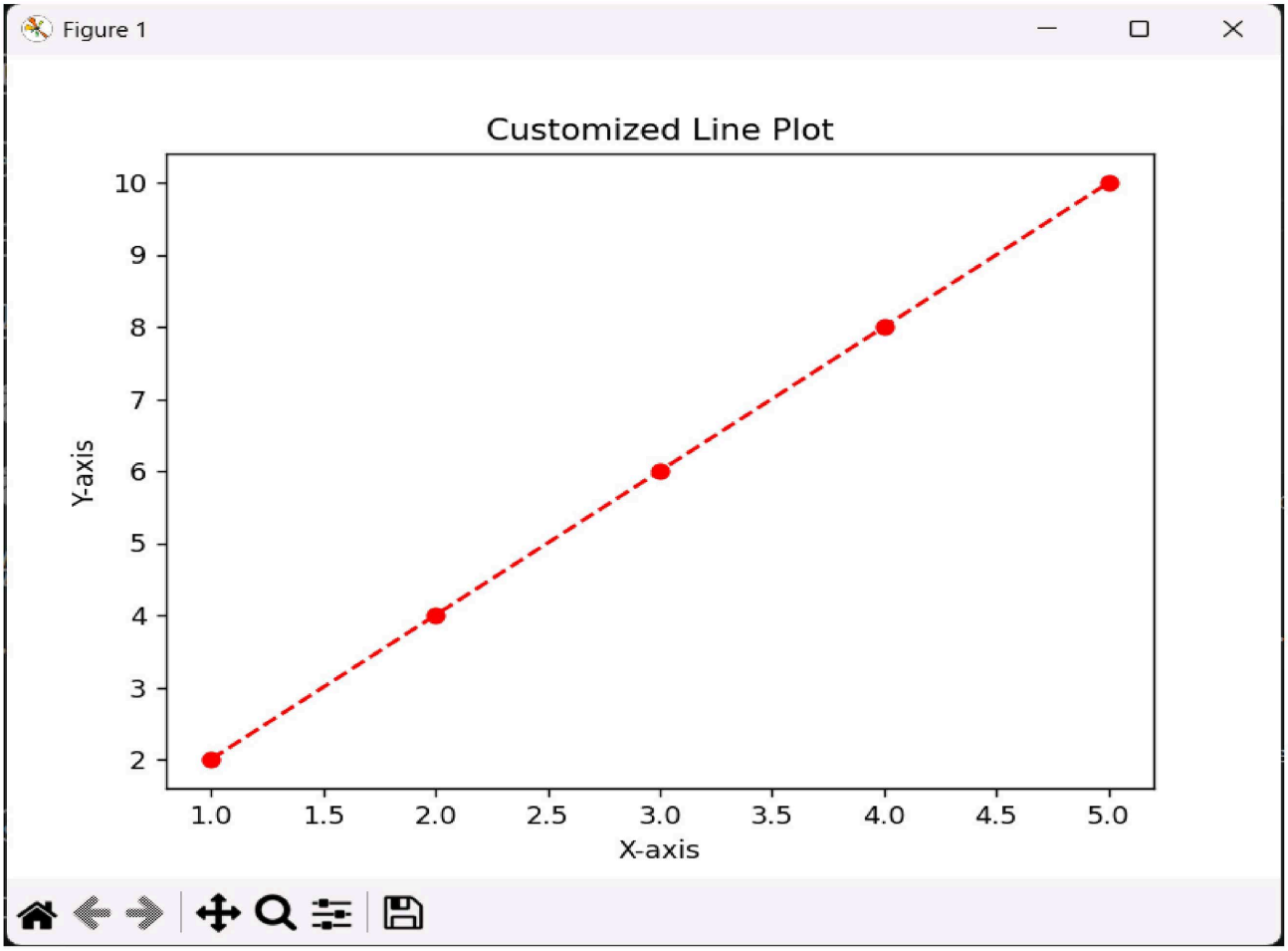
Figure 1
Customized Line Plot
Y-axis
X-axis
10
9
8
7
6
5
4
3
2
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

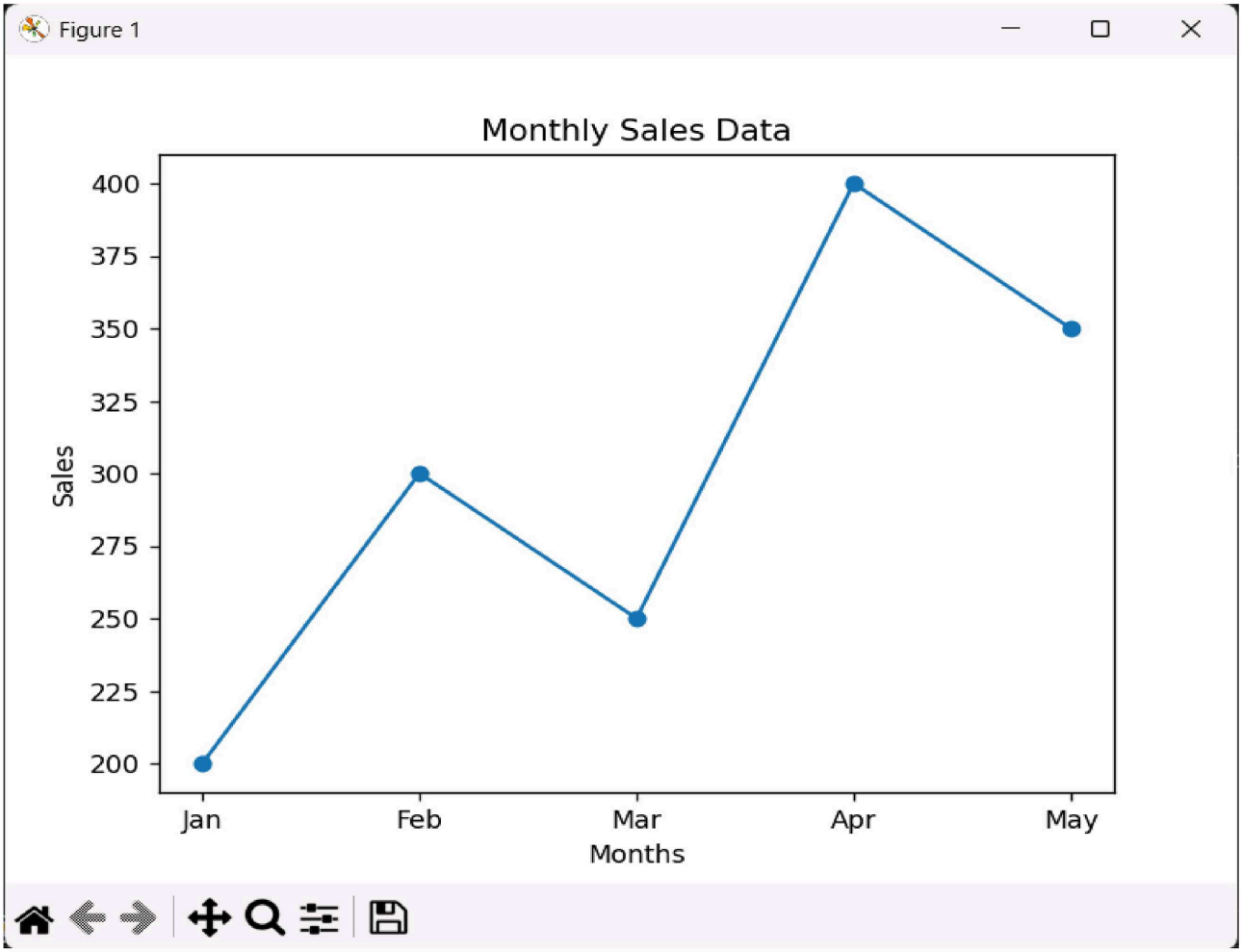
Figure 1
Monthly Sales Data
Sales
Months
400
375
350
325
300
275
250
225
200
Jan
Feb
Mar
Apr
May